Lab Report: RIPv2 Troubleshooting

Packet Tracer Activity 7.5.3

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This report documents the identification and remediation of routing misconfigurations across a multi-router Cisco Packet Tracer topology. The lab involved verifying IP reachability between branch offices and HQ, diagnosing missing RIP routing processes, version mismatches, and subnet route summarization issues across routers HQ, BRANCH1, and BRANCH2.

1 Network Topology Overview

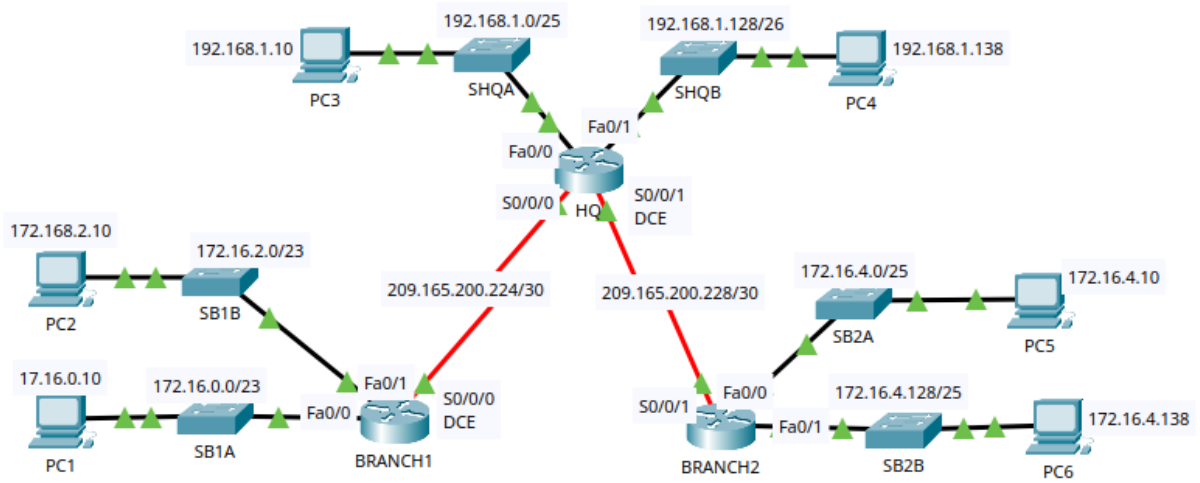


Figure 1: Overall Network Topology in Cisco Packet Tracer.

The network consists of three routers (*HQ, BRANCH1, BRANCH2*) connected via serial WAN links, with local area networks attached to each branch router.

2 Initial Troubleshooting: PC Connectivity Issues

2.1 Issue Identification

The first step involved verifying end-device connectivity. Testing from PC5 and PC6 revealed that they could not reach their respective default gateways.

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC1	BRANCH1	ICMP	
	Successful	PC3	HQ	ICMP	
	Successful	PC4	HQ	ICMP	
	Failed	PC5	BRANCH2	ICMP	
	Failed	PC6	BRANCH2	ICMP	

Figure 2: PC5 Failing to Reach Default Gateway.

```
Device Name: BRANCH2
Device Model: 1841
Hostname: BRANCH2

Port              Link   VLAN   IP Address
FastEthernet0/0   Up     --     172.16.4.129/25
FastEthernet0/1   Up     --     172.16.4.1/25
Serial0/0/0       Down   --     <not set>
Serial0/0/1       Up     --     209.165.200.230/30
Vlan1             Down   1      <not set>
```

Figure 3: BRANCH2's router interface configurations

2.2 Root Cause Analysis

Investigation of the router interface configurations revealed that the IP addresses on BRANCH2's FastEthernet interfaces were incorrectly assigned:

The IP addresses on the interfaces were swapped:

- **FastEthernet0/0:** Was configured as `172.16.4.129/25` but should be `172.16.4.1/25`
- **FastEthernet0/1:** Was configured as `172.16.4.1/25` but should be `172.16.4.129/25`

This misconfiguration meant that PCs on each subnet were trying to reach a gateway that belonged to the other subnet, resulting in failed connectivity.

2.3 Remediation

The interface IP addresses were swapped on BRANCH2:

```
BRANCH2# configure terminal
BRANCH2(config)# interface FastEthernet0/0
BRANCH2(config-if)# no ip address
BRANCH2(config-if)# ip address 172.16.4.1 255.255.255.128
BRANCH2(config-if)# no shutdown

BRANCH2(config)# interface FastEthernet0/1
BRANCH2(config-if)# no ip address
BRANCH2(config-if)# ip address 172.16.4.129 255.255.255.128
BRANCH2(config-if)# no shutdown
```

Listing 1: BRANCH2 Corrective Interface Configuration

2.4 Verification

After correcting the interface IP addresses, both PCs successfully reached their default gateways.

Fire	Last Status	Source	Destination	Type	Color
	Successful	PC5	BRANCH2	ICMP	
	Successful	PC6	BRANCH2	ICMP	
	Successful	PC5	PC6	ICMP	

Figure 4: PC5 Successfully Pinging Default Gateway After Fix.

3 Router HQ Configuration & Troubleshooting

3.1 Issue Identification

Analysis of HQ's RIP configuration revealed multiple issues:

1. **Empty RIP Routing Table:** The command `show ip route rip` returned no output, indicating that HQ was not learning any routes via RIP.
2. **Missing Network Statement for Branch Networks:** only `192.168.1.0` and `209.165.200.0` were listed. The `172.16.0.0` network was missing, meaning HQ was not advertising or learning the branch LAN subnets.
3. **Auto-Summary Enabled:** This caused VLSM subnets to be summarized to classful boundaries (`172.16.0.0/16`), preventing BRANCH1 and BRANCH2 from learning each other's specific subnets.

```
HQ#show ip route rip


HQ#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 4 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  Serial0/0/1           22
  Serial0/0/0           22
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
            192.168.1.0
            209.165.200.0
Passive Interface(s):
            FastEthernet0/0
            FastEthernet0/1
Routing Information Sources:
            Gateway         Distance      Last Update
Distance: (default is 120)
```

Figure 5: HQ RIP Protocols Configuration.

3.2 Remediation

The following configuration was applied to HQ:

```
HQ# configure terminal
HQ(config)# router rip
HQ(config-router)# version 2
HQ(config-router)# no auto-summary              ! CRITICAL: Preserve VLSM
HQ(config-router)# network 172.16.0.0           ! Was MISSING
HQ(config-router)# end
```

Listing 2: HQ Corrective RIP Configuration

4 Router BRANCH1 Configuration & Troubleshooting

4.1 Issue Identification

Inspection of BRANCH1's routing table revealed that it was learning only limited routes.

```
BRANCH1>en
BRANCH1#show ip route rip
     172.16.0.0/23 is subnetted, 2 subnets
R    192.168.1.0/24 [120/1] via 209.165.200.225, 00:00:11, Serial0/0/0
     209.165.200.0/30 is subnetted, 2 subnets
R       209.165.200.228 [120/1] via 209.165.200.225, 00:00:11, Serial0/0/0
```

Figure 6: BRANCH1 Pre-Fix RIP Routing Table.

The following problems were identified:

1. **Missing BRANCH2 LAN Subnets:** BRANCH1 could only see `192.168.1.0/24` (HQ LAN) and `209.165.200.228/30` (BRANCH2's serial network). The BRANCH2 LAN subnets (`172.16.4.0/25`, `172.16.4.128/25`) were not present in the routing table.
2. **Running RIPv1 Instead of RIPv2:** Under `Default version control:`, the output showed `send version 1, receive any version`. This caused subnet mask information to be lost, breaking VLSM support.

3. **Auto-Summary Enabled:** The output showed `Automatic network summarization is in effect`. This caused VLSM subnets to be summarized to classful boundaries, preventing BRANCH2's specific subnets from being learned.

4.2 Remediation

The following configuration was applied to BRANCH1:

```
BRANCH1# configure terminal
BRANCH1(config)# router rip
BRANCH1(config-router)# version 2              ! Enable RIPv2
BRANCH1(config-router)# no auto-summary        ! Preserve subnet masks
BRANCH1(config-router)# network 172.16.0.0     ! Advertise LAN subnets
BRANCH1(config-router)# no passive-interface Serial0/0/0
BRANCH1(config-router)# end
```

Listing 3: BRANCH1 Corrective RIP Configuration

5 Router BRANCH2 Configuration & Troubleshooting

5.1 Issue Identification

Inspection of BRANCH2's routing table revealed similar limited route learning.

```
BRANCH2#show ip route rip
     172.16.0.0/25 is subnetted, 2 subnets
R    192.168.1.0/24 [120/1] via 209.165.200.229, 00:00:16, Serial0/0/1
     209.165.200.0/30 is subnetted, 2 subnets
R       209.165.200.224 [120/1] via 209.165.200.229, 00:00:16, Serial0/0/1
```

Figure 7: BRANCH2 Pre-Fix RIP Routing Table.

The following problems were identified:

1. **Missing BRANCH1 LAN Subnets:** BRANCH2 could only see `192.168.1.0/24` (HQ LAN) and `209.165.200.224/30` (BRANCH1's serial network). The BRANCH1 LAN subnets (`172.16.0.0/23`, `172.16.2.0/23`) were not present in the routing table.

2. **Auto-Summary Enabled:** Auto-summary was active, causing VLSM subnets to be summarized to classful boundaries.

5.2 Remediation

The following configuration was applied to BRANCH2:

```
BRANCH2# configure terminal
BRANCH2(config)# router rip
BRANCH2(config-router)# no auto-summary        ! Preserve subnet masks
BRANCH2(config-router)# network 172.16.0.0     ! Advertise LAN subnets
BRANCH2(config-router)# no passive-interface Serial0/0/1
BRANCH2(config-router)# end
```

Listing 4: BRANCH2 Corrective RIP Configuration

6 Post-Fix Verification

After applying all configuration changes, routing tables were verified on each router to confirm proper RIP convergence.

6.1 HQ Routing Table (Post-Fix)

```
HQ>en
HQ#show ip route rip
     172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
R       172.16.0.0/23 [120/1] via 209.165.200.226, 00:00:23, Serial0/0/0
R       172.16.2.0/23 [120/1] via 209.165.200.226, 00:00:23, Serial0/0/0
R       172.16.4.0/25 [120/1] via 209.165.200.230, 00:00:06, Serial0/0/1
R       172.16.4.128/25 [120/1] via 209.165.200.230, 00:00:06, Serial0/0/1
```

Figure 8: HQ Converged RIP Routing Table.

HQ now learns all branch subnets: `172.16.0.0/23` and `172.16.2.0/23` from BRANCH1, and `172.16.4.0/25` and `172.16.4.128/25` from BRANCH2, all with metric 1.

6.2 BRANCH1 Routing Table (Post-Fix)

```
BRANCH1#show ip route rip
     172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
R       172.16.4.0/25 [120/2] via 209.165.200.225, 00:00:08, Serial0/0/0
R       172.16.4.128/25 [120/2] via 209.165.200.225, 00:00:08, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
R       192.168.1.0/25 [120/1] via 209.165.200.225, 00:00:08, Serial0/0/0
R       192.168.1.128/26 [120/1] via 209.165.200.225, 00:00:08, Serial0/0/0
     209.165.200.0/30 is subnetted, 2 subnets
R       209.165.200.228 [120/1] via 209.165.200.225, 00:00:08, Serial0/0/0
```

Figure 9: BRANCH1 Converged RIP Routing Table.

BRANCH1 now learns BRANCH2's LAN subnets (`172.16.4.0/25`, `172.16.4.128/25`) and HQ's LAN subnets (`192.168.1.0/25`, `192.168.1.128/26`) with metric 2.

6.3 BRANCH2 Routing Table (Post-Fix)

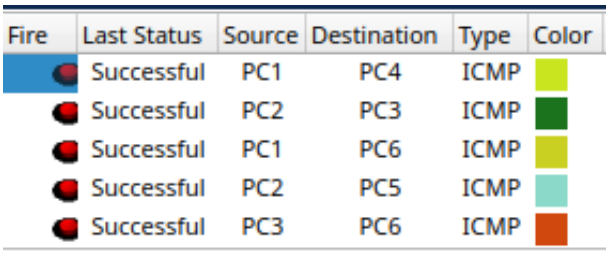
```
BRANCH2#show ip route rip
     172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
R       172.16.0.0/23 [120/2] via 209.165.200.229, 00:00:25, Serial0/0/1
R       172.16.2.0/23 [120/2] via 209.165.200.229, 00:00:25, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
R       192.168.1.0/25 [120/1] via 209.165.200.229, 00:00:25, Serial0/0/1
R       192.168.1.128/26 [120/1] via 209.165.200.229, 00:00:25, Serial0/0/1
     209.165.200.0/30 is subnetted, 2 subnets
R       209.165.200.224 [120/1] via 209.165.200.229, 00:00:25, Serial0/0/1
```

Figure 10: BRANCH2 Converged RIP Routing Table.

BRANCH2 now learns BRANCH1's LAN subnets (`172.16.0.0/23`, `172.16.2.0/23`) and HQ's LAN subnets (`192.168.1.0/25`, `192.168.1.128/26`) with metric 2.

7 Final End-to-End Verification

Final verification confirmed full end-to-end connectivity across the network:



Fire	Last Status	Source	Destination	Type	Color
	Successful	PC1	PC4	ICMP	
	Successful	PC2	PC3	ICMP	
	Successful	PC1	PC6	ICMP	
	Successful	PC2	PC5	ICMP	
	Successful	PC3	PC6	ICMP	

Figure 11: Final Verification Showing 100% Success Across All Endpoints.

All tests passed successfully, confirming full IPv4 connectivity across the topology.

8 Summary of Issues and Solutions

Router/Device	**Problem**	**Solution**
BRANCH2	Switched IP addresses on Fa0/0 and Fa0/1	Swapped IPs to correct interfaces
HQ	Missing `network 172.16.0.0` statement	Added `network 172.16.0.0`
HQ, BRANCH1, BRANCH2	Auto-summary enabled	Added `no auto-summary` on all routers
BRANCH1	Running RIPv1 instead of RIPv2	Added `version 2`

Table 1: Summary of All Issues and Their Resolutions.

9 Conclusion

This lab demonstrated the importance of proper RIP configuration in a multi-router network. The key lessons learned are:

1. **Always use RIPv2:** RIPv1 does not support VLSM, causing subnet mask information to be lost.
2. **Always disable auto-summary:** The `no auto-summary` command is essential to preserve subnet boundaries and enable CIDR.
3. **Verify network statements:** Each router must include `network` statements for all directly connected subnets that should participate in RIP.
4. **Check passive interfaces:** Serial WAN links must have RIP enabled with `no passive-interface`, while LAN interfaces should remain passive for security.